Econometrics (25117): Midterm 1, 2026–2027

Worked answer key, explanations and lecture references: 150-minute midterm

Monday, 9 November 2026

The maximum is 100 points; divide the raw total by 10 for the grade out of 10. Award method credit for correct reasoning with an arithmetic error, and avoid penalizing the same carried-forward error twice. Accept equivalent notation, exact fractions and reasonable rounding. The solutions below use the large-sample standard normal approximation, not an exact Student t reference.

Each answer below includes the lecture sentences or formulas used to solve it. Page numbers count PDF pages from 1, including the title page; they coincide with the printed slide numbers in Lectures 1–3. The page references are clickable links to the course PDFs. Keep this key in Exams/, with the lecture PDFs in the parent course folder and seminar PDFs in seminars/ under that parent folder.

Quotation marks identify lecture wording. A sentence marked "Application" explains a calculation or deduction for this exam; it is not presented as a quotation from the slides. Supplemental seminar references identify matching practice where useful. The exam data differ from the examples in the lectures.

Multiple-choice questions

Question	1	2	3	4	5	6	7	8
Answer	b	c	b	a	d	b	d	c

Four points per correct option; zero for a wrong or blank response.

1. Answer (b). $\mathrm{E}[Y_1] = \mu$, so the estimator is unbiased. Its distribution never concentrates at μ as n grows: the same nondegenerate random variable is used for every n. In particular, for some $\varepsilon > 0$, $\Pr(|Y_1 - \mu| > \varepsilon) > 0$ remains unchanged. Thus it is inconsistent. The positive variance ensures nondegeneracy. Option (a) incorrectly claims consistency; (c) and (d) incorrectly claim bias, and (c) also incorrectly claims consistency. Merely observing a nonvanishing variance is not a general proof of inconsistency for arbitrary estimator sequences; here the unchanged distribution supplies the argument.

   Lecture 1, PDF p. 21, "Random Sampling":

   "Then, the expected value of every draw is the population mean and the variance of every draw is the population variance."

   Lecture 1, PDF p. 27, "Estimators":

   "$\hat{\mu}_Y$ is *unbiased* if $\mathbb{E}(\hat{\mu}_Y) = \mu_Y$"

   The same slide defines consistency by $\hat{\mu}_Y \xrightarrow{p} \mu_Y$. Application: substitute Y_1 into these definitions. Its expectation is right, but its unchanged distribution does not concentrate at μ as n grows. This particular counterexample is an application of the definitions, rather than a worked numerical example on that slide.

2. Answer (c). Independence implies $\mathrm{E}[XY] = \mathrm{E}[X]\,\mathrm{E}[Y]$ and hence zero covariance. Options (a) and (b) reverse implications that do not hold in general. For example, let X be uniform on $[-1, 1]$ and $Y = X^2$. Symmetry gives $\mathrm{Cov}(X, Y) = 0$, but Y depends on X and $\mathrm{E}[Y \mid X] = X^2$ is not constant. Option (d) is false because perfect positive correlation between nondegenerate variables means an exact increasing linear relationship, not independence.

   Lecture 1, PDF p. 19, "Covariance":

   "When X and Y are independent, $\sigma_{XY} = 0$. The converse is not true in general: zero covariance does not necessarily imply independence."

   The quadratic plot in panel (d) of Lecture 1, PDF p. 20 illustrates a nonlinear relationship with zero correlation. Application: the $Y = X^2$ example above also shows why zero covariance need not make the conditional mean constant. The uniform-distribution calculation is an added illustration of the slide's point, not a quotation of a worked lecture example.

3. Answer (b). The CLT is $\sqrt{n}(\bar{Y}-\mu)/\sigma \xrightarrow{d} N(0,1)$. The factor $\sqrt{n}$ offsets the shrinking spread of the sample mean. Option (a) incorrectly makes an exact finite-sample normality claim without assuming a normal population. In (c), the expression without $\sqrt{n}$ converges in probability to zero. Option (d) is false because the law of large numbers gives $\bar{Y} \xrightarrow{p} \mu$, not necessarily zero.

   Lecture 1, PDF p. 25, "Asymptotic properties ($n \to \infty$) of the sampling distribution (2/2)":

   "If $Y_1, Y_2, \ldots, Y_n$ are drawn randomly from the same population (i.i.d.) and large outliers are unlikely, then $\overline{Y}$ is approximately distributed as $\mathcal{N}(\mu_Y, \sigma_Y^2/n)$ as n grows"

   The CLT box on this page states

$$\frac{\overline{Y}-\mu_Y}{\sigma_Y/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1).$$

   Lecture 1, PDF p. 25, "Asymptotic properties ($n \to \infty$) of the sampling distribution (2/2)":

   "$\overline{Y} \sim \mathcal{N}(\mu_Y, \sigma_{\overline{Y}}^2)$ exactly when $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, and approximately when Y follows another distribution"

   Lecture 1, PDF p. 23, "Asymptotic properties ($n \to \infty$) of the sampling distribution (1/2)":

   "In this case, we say that $\overline{Y}$ converges in probability to $\mathbb{E}(Y)$"

   Application: the exam does not assume a normal population, so the exact finite-sample claim in (a) is not guaranteed. The CLT formula is algebraically the same as (b). Convergence in probability concerns getting arbitrarily close to the mean: $\Pr(|\bar{Y}-\mu| > \varepsilon) \to 0$ for every $\varepsilon > 0$; it does not require exact equality.

4. Answer (a). A Type I error rejects a true null. Option (b) describes a Type II error: failing to reject a false null. Options (c) and (d) describe correct decisions, so neither is an error.

   Lecture 2, PDF p. 14, "Type I and Type II Errors in Hypothesis Testing":

   "Occurs when we reject a null hypothesis that is actually true."

   This is the sentence in the Type I box. The Type II box on the same page says: "Occurs when we fail to reject a null hypothesis that is actually false." Application: these are exactly the descriptions in options (a) and (b), respectively; only the first answers the Type I question.

5. Answer (d). Coverage is a repeated-sampling property of the interval procedure. In the frequentist framework the parameter is fixed, and the interval is random before the sample is observed. With the approximation used here, coverage is approximately 95%. Option (a) assigns a probability to the fixed parameter after observing the interval. Option (b) confuses uncertainty about a mean with dispersion of individual observations. Option (c) confuses interval coverage with exact equality between sample and population means.

   Lecture 2, PDF p. 20, "What are Confidence Intervals (CIs)?":

   "They provide a range of plausible values for a population parameter, not just a single point estimate."

   Lecture 2, PDF p. 21, "Computing CIs":

   "For $Z \sim \mathcal{N}(0,1)$, choose c so that $P(-c \le Z \le c) = 1-\alpha$."

   The slide gives the interval $\overline{y} \pm c \times SE(\overline{Y})$. Application: with $c = 1.96$, the large-sample probability statement can be rewritten as

$$\Pr\Big(\mu_Y \in [\overline{Y} - 1.96SE(\overline{Y}),\ \overline{Y} + 1.96SE(\overline{Y})]\Big) \approx 0.95.$$

   The interval varies across samples while μ_Y stays fixed. This explains the repeated-sampling wording in (d). The slides give the probability construction; they do not print that exact repeated-sampling sentence.

6. Answer (b). The two residual identities are the sample OLS first-order conditions when an intercept is included, regardless of the population relationship between X_i and U_i. Option (a) confuses these mechanical sample identities with the population condition $\mathrm{E}[U_i \mid X_i] = 0$. Option (c) equates the unobserved population error with

the fitted residual, which generally differ because coefficients are estimated. Option (d) would require a perfect fit; OLS makes residuals sum to zero, not each residual equal zero.

Lecture 3, PDF p. 7, "The Ordinary Least Squares (OLS) Estimator":

"The first-order conditions are"

$$\sum_{i=1}^{n}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \qquad \sum_{i=1}^{n}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)X_i = 0.$$

Using $\hat{U}_i = Y_i - \hat{Y}_i$ from Lecture 3, PDF p. 9 gives the two residual identities in option (b).

Lecture 3, PDF p. 19, "The conditions for causality":

"The error term U_i has conditional mean 0 given X_i: $\mathbb{E}[U_i \mid X_i] = 0$;"

Application: this is a population restriction, separate from the sample equations used to fit the line.

Supplemental practice: Seminar 3, PDF p. 11, "Solution 3: Normal equations and squared errors (continued)":

"These sample fitting identities do not establish causality."

7. Answer (d). At $n = 100$, the standard deviation of the sample mean is $\sigma_Y/10$; at $n = 400$, it is $\sigma_Y/20$. The ratio is $1/2$, so the standard deviation is halved. Option (a) gives the change in the variance, which is divided by four. Options (b) and (c) incorrectly say that uncertainty increases or stays unchanged as the sample size quadruples.

   Lecture 1, PDF p. 22, "Sample Distribution Mean and Dispersion":

   "The standard deviation of the sampling distribution is therefore given by $\frac{\sigma_Y}{\sqrt{n}}$"

   Application: the ratio of the new standard deviation to the old one is $\sqrt{100}/\sqrt{400} = 10/20 = 1/2$. The population distribution has not changed; the sample mean has become more precise.

8. Answer (c). $SER^2 = SSR/(n-2)$ accounts for the two fitted coefficients, while $RMSE^2 = SSR/n$. Both describe the size of fitted residuals; neither is the standard error of the slope. Option (a) reverses the denominators. Option (b) is false because $SER > RMSE$ for positive SSR and finite $n > 2$. Option (d) uses $n - 1$, the denominator for a sample variance around an estimated mean, rather than $n-2$ for this regression or n for $RMSE$.

   Lecture 3, PDF p. 16, "SER and RMSE: What Is the Difference?":

   "Both measures summarize the typical size of the OLS residuals."

   The two displayed definitions on that page are

$$SER = \sqrt{\frac{SSR}{n-2}}, \qquad RMSE = \sqrt{\frac{SSR}{n}}.$$

   The slide explains: "The two degrees of freedom are used to estimate β_0 and β_1." Application: these are the formulas in (c); for the same positive SSR, dividing by the smaller denominator makes SER larger than $RMSE$.

Long-answer questions

Question 9: joint probabilities and dependence (10 points)

(a) $\Pr(D = 1) = 0.24 + 0.36 = 0.60$ and $\Pr(Y = 1) = 0.04 + 0.36 = 0.40$. Each marginal probability sums the joint cells over the other variable. Award 1 point each. The "Marginal Probability distribution" formula on Lecture 1, PDF p. 13, "Joint Probability Distribution $\Pr(X = x, Y = y)$", is

$$\Pr(Y = y) = \sum_i \Pr(X = x_i, Y = y).$$

Application: sum down a column for the marginal of D and across a row for the marginal of Y. The result uses the exam's cells, not the lecture's different joint-distribution example.

(b) Because Y is binary,

$$\mathrm{E}[Y \mid D = 0] = \frac{0.04}{0.40} = 0.10, \qquad \mathrm{E}[Y \mid D = 1] = \frac{0.36}{0.60} = 0.60.$$

For a zero–one variable, its mean is the probability of being one; the denominator restricts attention to the group being conditioned on. The law of iterated expectations then gives $\mathrm{E}[Y] = 0.10(0.40) + 0.60(0.60) = 0.40$. The overall mean is a weighted average of the two group means, with the population group shares as weights. Award 1 point for each conditional mean and 1 point for the weighted average. The conditional-probability formula on Lecture 1, PDF p. 13 is $\Pr(Y = y \mid X = x) = \Pr(X = x, Y = y)/\Pr(X = x)$.

Lecture 1, PDF p. 14, "Working with the joint distribution":

"Same cell, different denominator: conditioning changes the reference group."

Lecture 1, PDF p. 16, "Calculating the law of iterated expectations":

"Because Y is binary, $\mathbb{E}(Y \mid X = x) = \Pr(Y = 1 \mid X = x)$."

The "Law of Iterated Expectations" formula on Lecture 1, PDF p. 15 is

$$\mathbb{E}(Y) = \mathbb{E}[\mathbb{E}(Y \mid X)] \equiv \sum_i \mathbb{E}(Y \mid X = x_i) \Pr(X = x_i).$$

Application: replace X by D, calculate the two within-group probabilities, and weight them by the population shares 0.40 and 0.60.

(c) The product DY equals one only when both variables equal one. Thus $\mathrm{E}[DY] = \Pr(D = 1, Y = 1) = 0.36$, so

$$\mathrm{Cov}(D, Y) = 0.36 - (0.60)(0.40) = 0.12.$$

For a binary variable with probability p of being one, its variance is $p(1 - p)$. Both variances therefore equal $0.60(0.40) = 0.24$. Consequently,

$$\mathrm{Corr}(D, Y) = \frac{0.12}{\sqrt{0.24 \cdot 0.24}} = 0.50.$$

Correlation divides covariance by the product of the two standard deviations, removing the units. The positive sign means that workshop attendance and employment are positively associated; it does not by itself establish a causal effect. Award 1.5 points for the covariance and 1.5 for the correlation, including the variance calculation.

Lecture 1, PDF p. 19, "Covariance":

"The covariance between X and Y, σ_{XY} or $\mathrm{Cov}(X, Y)$, is the average product of how two variables differ from their means. It measures how X and Y move together."

Lecture 1, PDF p. 20, "Correlation":

"To clarify the relationship, covariance is normalized by the variables' standard deviations."

The correlation formula on that slide is $\rho_{XY} = \sigma_{XY}/(\sigma_X \sigma_Y) \in [-1, 1]$. The moment definitions on Lecture 1, PDF p. 10, "Statistical Moments", include $\mathrm{Var}(Y) = \mathbb{E}[(Y - \mu_Y)^2]$. Application: expanding the covariance definition gives $\mathrm{Cov}(D, Y) = \mathrm{E}[DY] - \mathrm{E}[D]\,\mathrm{E}[Y]$. For a zero–one variable B, $B^2 = B$, so its variance is $\mathrm{E}[B^2] - \mathrm{E}[B]^2 = p - p^2 = p(1-p)$. These algebraic steps give the covariance and both variances used above.

(d) They are not independent: $\Pr(D = 1, Y = 1) = 0.36$ differs from $\Pr(D = 1)\Pr(Y = 1) = 0.24$. Equivalently, $\Pr(Y = 1 \mid D = 1) = 0.60 \neq 0.40 = \Pr(Y = 1)$. Award 1 point for the conclusion and 1 point for a valid probability comparison.

Lecture 1, PDF p. 17, "Independence":

"The joint probability distribution of X and Y can be expressed as the product of their individual probability distributions"

The same slide writes the independence criterion as

$$\Pr(Y = y \mid X = x) = \Pr(Y = y) \quad \Longleftrightarrow \quad \Pr(X = x, Y = y) = \Pr(Y = y)\Pr(X = x).$$

Application: one cell violating the product equality is enough to rule out independence. Here $0.36 \neq 0.24$.

Question 10: comparing two means (11 points)

(a) The estimated difference is $\widehat{\Delta} = 74 - 68 = 6$ points. Under the stated independence,

$$SE(\widehat{\Delta}) = \sqrt{\frac{12^2}{100} + \frac{16^2}{100}} = \sqrt{1.44 + 2.56} = 2 \text{ points.}$$

Each sample mean has estimated variance s_g^2/n_g. Independence makes the variance of their difference the sum of these variances; the standard error is the square root of that sum. Award 1 point for the estimate and 1 point for the standard error.

Lecture 2, PDF p. 22, "Mean Difference Testing: Comparing Groups 1/2":

"Assume two independent samples of sizes n_a and n_b, with i.i.d. observations within each sample. Thus $\mathrm{Cov}(\overline{Y}^a, \overline{Y}^b) = 0$."

The slide says: "Estimate the standard error using the sample standard deviations, s_Y^a and s_Y^b:" and gives

$$SE(\overline{Y}^a - \overline{Y}^b - d_0) = \sqrt{\frac{(s_Y^a)^2}{n_a} + \frac{(s_Y^b)^2}{n_b}}.$$

Application: subtracting the fixed null value d_0 does not change the variance. Insert the two sample SDs and sizes to obtain $SE = 2$.

(b) Test $H_0 : \Delta = 2$ against $H_1 : \Delta \neq 2$. The observed test statistic is $t_{\text{obs}} = (6 - 2)/2 = 2$. Award 0.5 points for each hypothesis and 2 points for the correctly centered and standardized statistic. Using $6/2 = 3$ tests the wrong null.

Lecture 2, PDF p. 23, "Mean Difference Testing: Comparing Groups 2/2":

"Once we have $SE(\overline{Y}^a - \overline{Y}^b)$, we can easily compute the corresponding t-statistic"

$$t = \frac{(\overline{y}^a - \overline{y}^b) - d_0}{SE(\overline{Y}^a - \overline{Y}^b)}.$$

The hypotheses on Lecture 2, PDF p. 22 allow a general null difference d_0. Application: set $d_0 = 2$ here. The numerator is the distance between the estimated difference 6 and the claimed difference 2.

(c) For the two-sided test, compare the absolute statistic with each critical value. At 5%, $|2| > 1.96$, so reject H_0. At 1%, $|2| < 2.576$, so do not reject H_0. A stricter significance level requires stronger evidence against the null. Award 1 point for each comparison and its correct decision. No probability-table lookup is required.

Lecture 2, PDF p. 23, "Mean Difference Testing: Comparing Groups 2/2":

"We can reject that $\mu_Y^a - \mu_Y^b = d_0$ at the 5%-level when $\mid t \mid > 1.96$"

Supplemental practice: Seminar 2, PDF p. 10, "Solution 2: Decisions and the meaning of .0164 (continued)":

"Reject H_0 when the p-value is at most α, or equivalently when $|t^{\text{obs}}|$ reaches the two-sided critical value."

The seminar table prints critical values 1.960 at 5% and 2.576 at 1%. The table on Lecture 2, PDF p. 21 prints the 1% value rounded to 2.58. Application: use the more precise value supplied in the exam; either rounding gives the same decision for $|t| = 2$.

(d) The 95% interval is

$$6 \pm 1.96(2) = [2.08, 9.92] \text{ points.}$$

It excludes the null value 2, consistently with rejection by the two-sided 5% test using the same normal approximation. Award 0.5 points for the formula, 0.5 for each endpoint, 0.25 for checking the value 2, and 0.25 for the test–interval connection. The interval concerns the population difference, not the spread of individual scores.

Lecture 2, PDF p. 23, "Mean Difference Testing: Comparing Groups 2/2":

"Similarly, we can construct the confidence interval for $\mu_Y^a - \mu_Y^b$"

$$CI_{\mu_Y^a - \mu_Y^b} = (\overline{y}^a - \overline{y}^b) \pm 1.96 \times SE(\overline{Y}^a - \overline{Y}^b).$$

The same slide states: "We can reject that $\mu_Y^a - \mu_Y^b = d_0$ at the 5%-level when $d_0 \notin CI_{\mu_Y^a - \mu_Y^b}$." Application: the excluded value is 2, which is the null being tested; checking only whether zero belongs to the interval would answer a different test question.

(e) With m students in each independent group, the planned standard error is $\sqrt{12^2/m + 16^2/m} = 20/\sqrt{m}$. The planned 95% interval half-width is therefore

$$h(m) = 1.96\frac{20}{\sqrt{m}} = \frac{39.2}{\sqrt{m}}.$$

Requiring $h(m) \leq 2$ gives $\sqrt{m} \geq 19.6$, or $m \geq 19.6^2 = 384.16$. Sample size must be an integer and must meet the inequality, so round up: $m_{\min} = 385$ students per group, or 770 in total. For comparison, $m = 384$ gives a half-width just above 2. These are planning calculations using the supplied standard deviations; the realized interval width can differ when standard deviations are estimated from the new data. Award 1 point for the half-width inequality and bound, 0.5 for rounding up to 385 per group, and 0.5 for the total of 770. Application of the independent-group variance formula on Lecture 2, PDF p. 22 and the interval formula on Lecture 2, PDF p. 23: replace both group sizes by m and solve the half-width inequality. This two-group planning calculation combines the lecture formulas; it is not a verbatim worked example in the lecture.

Supplemental practice: Seminar 2, PDF p. 20, "Solution 6: Sample size for a target half-width":

"For $m_n \leq m$, solve and round up:"

That seminar uses m for a target half-width and n for the sample size, and gives $n_{\min} = \lceil (1.96\sigma/m)^2 \rceil$ for one mean. In this exam, m instead denotes students per group. Apply the same solve-and-round-up method to the two-group standard error derived above. The seminar also states: "Realized precision depends on how well the planning value describes the population."

Question 11: simple OLS (11 points)

(a) For simple OLS with an intercept, divide the centered cross-product sum by the centered sum of squares of X:

$$\widehat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{50}{100} = 0.5, \qquad \widehat{\beta}_0 = 70 - 0.5(4) = 68.$$

The intercept follows from the fact that the fitted line passes through $(\bar{x}, \bar{y})$. The fitted line is $\widehat{y}_i = 68 + 0.5x_i$: one additional weekly study hour is associated with 0.5 additional predicted score points. This is not yet a causal statement. Award 0.5 points each for the slope, intercept, fitted line and interpretation.

Lecture 3, PDF p. 8, "The Ordinary Least Squares (OLS) Estimator":

"The intercept adjusts the average outcome for the fitted slope."

The displayed coefficient formulas are

$$\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}, \qquad \hat{\beta}_1 = \frac{\sum_i (Y_i - \overline{Y})(X_i - \overline{X})}{\sum_i (X_i - \overline{X})^2} = \frac{s_{XY}}{s_X^2}.$$

The same slide says: "The slope compares the comovement of X and Y with the variation in X."

Lecture 3, PDF p. 3, "The regression line":

"The slope is the difference in the expected values of Y, for two values of X that differ by one unit"

Application: use the supplied centered sums in the estimator formula and read the fitted slope as the change in predicted score per additional hour. The estimated-line interpretation alone makes no causal claim.

(b) $\widehat{y}_i = 68 + 0.5(6) = 71$ and $\widehat{u}_i = 72 - 71 = 1$. The fitted score is the point on the estimated line at six hours. The residual is observed minus fitted, so this student scores one point above the prediction. Award 0.5 points for the fitted score and 0.5 for the residual. The formulas on Lecture 3, PDF p. 9, "Back to the Linear Regression Model", are

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i, \qquad \hat{U}_i = Y_i - \hat{Y}_i.$$

Supplemental practice: Seminar 3, PDF p. 10, "Solution 3: Fitted scores and vertical residuals":

"Residuals use the estimated sample line. Population errors use the true population coefficients."

Application: substitute six hours into the fitted line, then subtract its prediction from the observed score. Reversing the subtraction would give the wrong residual sign.

(c) Since OLS includes an intercept,

$$ESS = \widehat{\beta}_1^2 \sum_i (x_i - \bar{x})^2 = 0.5^2(100) = 25, \qquad SSR = TSS_Y - ESS = 425 - 25 = 400.$$

Thus $R^2 = 25/425 = 1/17 \approx 0.0588$ and $SER = \sqrt{400/(102-2)} = 2$ score points. The fitted values account for about 5.88% of the sample variation in scores. SER summarizes residual dispersion in score points, using $n - 2$ because the fitted line estimates an intercept and a slope. It is not a coefficient standard error. Award 1 point each for ESS, SSR, R^2 and SER.

Lecture 3, PDF p. 13, "R^2":

"The R^2 is the ratio of the sample variance of $\hat{Y}$ to the sample variance of Y"

This page writes R^2 as the explained sum of squares divided by the total sum of squares, and also as one minus the residual sum of squares divided by the total sum of squares.

Lecture 3, PDF p. 12, "Goodness of Fit":

"The standard error of the regression (SER) measures the magnitude of a typical regression residual in the units of Y."

The definition $SER = \sqrt{SSR/(n-2)}$ appears on Lecture 3, PDF p. 16.

Supplemental practice: Seminar 3, PDF p. 19, "Solution 6: Decomposing the observed score variation":

"Thus $TSS = ESS + SSR$: $32 = 25.6 + 6.4$."

Application: the seminar numbers illustrate the decomposition. For this exam, $\hat{y}_i - \bar{y} = 0.5(x_i - \bar{x})$, so $ESS = 0.5^2(100) = 25$ and the corresponding decomposition is $425 = 25 + 400$.

(d) Sample correlation can be calculated directly from centered sums:

$$r_{XY} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}} = \frac{50}{\sqrt{100 \cdot 425}} = \frac{1}{\sqrt{17}} \approx 0.2425.$$

The sample covariance and both sample variances would each use the same $n-1$ divisor; it cancels in this correlation formula. Squaring gives $r_{XY}^2 = 1/17 \approx 0.0588 = R^2$. This equality holds for simple OLS with an intercept. Correlation retains the positive sign of the association, whereas R^2 records its squared strength. Award 1 point for the correlation and 1 point for verifying the equality numerically.

Lecture 2, PDF p. 28, "The California Test Score Data Set":

"The sample correlation, $r_{XY} = \frac{s_{XY}}{s_X s_Y}$"

The same page gives $s_{XY} = (n-1)^{-1} \sum_i (X_i - \overline{X})(Y_i - \overline{Y})$. Application: the $n-1$ terms cancel when this covariance is divided by the two sample standard deviations, giving the centered-sum correlation above.

The identity $r^2 = R^2$ is an algebraic consequence of the OLS formulas on Lecture 3, PDF p. 8 and the fit definition on Lecture 3, PDF p. 13; it is not printed there as a separate sentence. Writing S_{XX}, S_{XY}, S_{YY} for the three centered sums supplied in the question gives

$$ESS = \hat{\beta}_1^2 S_{XX} = \frac{S_{XY}^2}{S_{XX}}, \qquad R^2 = \frac{S_{XY}^2}{S_{XX} S_{YY}} = r_{XY}^2.$$

This derivation requires simple OLS with an intercept, as in the exam.

(e) Substitute $X_i = M_i/60$ into the original fitted line:

$$\widehat{y}_i = 68 + 0.5\left(\frac{M_i}{60}\right) = 68 + \frac{1}{120} M_i.$$

The intercept remains 68, and the slope is $1/120 \approx 0.00833$ score points per additional study minute. A minute is $1/60$ of an hour, so its slope is $1/60$ of the hourly slope. The conversion changes the numerical regressor units, but it does not change any fitted score or residual. Consequently ESS, SSR and TSS_Y remain the same, giving $R^2 = 1/17 \approx 0.0588$ and $SER = 2$ score points again. Award 0.5 points each for the intercept, slope, unchanged R^2 and unchanged SER. Application of the fitted-value formula on Lecture 3, PDF p. 9: expressing the same study time in minutes means $X_i = M_i/60$. Substitute that identity without changing the fitted score.

Supplemental practice: Seminar 3, PDF p. 13, "Solution 4: Shares, percentage points, and predicted scores":

"Changing the units changes the coefficient's numerical scale and preserves the predictions."

The seminar example converts a share to percentage points; the exam applies the same substitution method to hours and minutes.

Lecture 3, PDF p. 16, "SER and RMSE: What Is the Difference?":

"Both are measured in the units of Y. They describe the typical size of the part of Y that the regression does not explain."

Application: the outcome units and every residual are unchanged. Therefore the residual sum of squares and SER are unchanged. The explained and total sums of squares are also unchanged, so R^2 stays the same, consistently with its unitless interpretation on Lecture 3, PDF p. 12.

Additional paper-and-pencil problems: Questions 12–14

Question 12: Lesson 1—a small library population (12 points)

(a) There is one student with $Y = 0$, six with $Y = 2$ and one with $Y = 4$. Each student has probability $1/8$ of being selected, so

$$\Pr(Y = 0) = \frac{1}{8}, \qquad \Pr(Y = 2) = \frac{6}{8} = \frac{3}{4}, \qquad \Pr(Y = 4) = \frac{1}{8}.$$

The probabilities add to one. Accumulating the probabilities from left to right gives the complete c.d.f.:

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ 1/8, & 0 \le y < 2, \\ 7/8, & 2 \le y < 4, \\ 1, & y \ge 4. \end{cases}$$

Award 1.5 points for the marginal distribution and 1.5 points for the complete c.d.f., including its values at the jumps.

Lecture 1, PDF p. 7, "Some Revision":

"The cumulative distribution function (c.d.f.) is the probability that the random variable is less than or equal to a particular value"

The marginal-distribution formula on Lecture 1, PDF p. 13, "Joint Probability Distribution $\Pr(X = x, Y = y)$", is $\Pr(Y = y) = \sum_i \Pr(X = x_i, Y = y)$. Application: add the two programme counts in each row and divide by eight, then sum the probabilities at all outcomes no larger than y.

Supplemental practice: Seminar 1, PDF p. 5, "Solution 1: The earnings c.d.f. (continued)":

"At each jump, the filled dot gives the value of the right-continuous c.d.f."

Application: use the same step-function construction, with this problem's jumps at 0, 2 and 4. In particular, $F_Y(0) = 1/8$, $F_Y(2) = 7/8$ and $F_Y(4) = 1$.

(b) Weight each value and its square by its probability:

$$\mathrm{E}[Y] = 0\left(\frac{1}{8}\right) + 2\left(\frac{3}{4}\right) + 4\left(\frac{1}{8}\right) = \frac{3}{2} + \frac{1}{2} = 2,$$

$$\mathrm{E}[Y^2] = 0^2\left(\frac{1}{8}\right) + 2^2\left(\frac{3}{4}\right) + 4^2\left(\frac{1}{8}\right) = 3 + 2 = 5.$$

Therefore

$$\mathrm{Var}(Y) = \mathrm{E}[Y^2] - (\mathrm{E}[Y])^2 = 5 - 2^2 = 1.$$

A direct check gives $\mathrm{Var}(Y) = (0-2)^2/8 + (2-2)^2(3/4) + (4-2)^2/8 = 1$. The mean is two books and the variance is one book squared. These are moments of the specified eight-student population, not estimates from a sample; there is no $n - 1$ correction. Award 1 point for each requested moment.

Lecture 1, PDF p. 10, "Statistical Moments":

"The mean (or expected value) indicates the long-run average of a random variable Y:"

That slide gives $\mathbb{E}(Y) = \sum_i y_i \Pr(Y = y_i)$ and $\mathrm{Var}(Y) = \mathbb{E}[(Y - \mu_Y)^2]$. The variance shortcut on Lecture 1, PDF p. 11, "Calculating variance two ways", is $\mathrm{Var}(Y) = \mathbb{E}(Y^2) - [\mathbb{E}(Y)]^2$. Application: substitute the marginal probabilities from part (a).

(c) Each programme contains four students, so $\Pr(G = \mathrm{A}) = \Pr(G = \mathrm{B}) = 4/8 = 1/2$. Programme A contains one student borrowing zero books, two borrowing two and one borrowing four. Programme B has four students, all borrowing two. Thus

$$\mathrm{E}[Y \mid G = \mathrm{A}] = \frac{0(1) + 2(2) + 4(1)}{4} = \frac{8}{4} = 2, \qquad \mathrm{E}[Y \mid G = \mathrm{B}] = \frac{2(4)}{4} = 2.$$

Iterated expectations weights the two conditional means by their population programme shares:

$$\mathrm{E}[Y] = \mathrm{E}[Y \mid G = \mathrm{A}] \Pr(G = \mathrm{A}) + \mathrm{E}[Y \mid G = \mathrm{B}] \Pr(G = \mathrm{B}) = 2\left(\frac{1}{2}\right) + 2\left(\frac{1}{2}\right) = 2.$$

This agrees with part (b). Award 1 point for each conditional mean and 1 point for the correctly weighted recovery of the marginal mean. The formulas on Lecture 1, PDF p. 15, "Conditional and Iterated Expectations", are

$$\mathbb{E}(Y \mid X = x) = \sum_i y_i \Pr(Y = y_i \mid X = x), \qquad \mathbb{E}(Y) = \sum_i \mathbb{E}(Y \mid X = x_i) \Pr(X = x_i).$$

Lecture 1, PDF p. 14, "Working with the joint distribution":

"Same cell, different denominator: conditioning changes the reference group."

Application: within a programme, divide its counts by four rather than eight. For the marginal mean, average the two programme means using their shares.

(d) The variables are not independent. For example,

$$\Pr(Y = 2 \mid G = \mathrm{B}) = \frac{4}{4} = 1, \qquad \Pr(Y = 2) = \frac{6}{8} = \frac{3}{4}.$$

Knowing that the selected student is in programme B changes the probability of borrowing two books. The conditional means happen to be equal, but the conditional distributions differ: programme A includes zero and four books, whereas programme B contains only two. Equality of conditional means alone therefore does not imply independence. Award 1.5 points for the conditional-versus-marginal probability comparison and 1.5 points for the conclusion about independence and equal means. The independence criterion on Lecture 1, PDF p. 17, "Independence", includes $\Pr(Y = y \mid X = x) = \Pr(Y = y)$. Application: the displayed failure at $y = 2$ rules out independence. The conclusion about equal means is a deduction from this problem's calculations, not a quotation of a worked example in the lecture.

Question 13: Lesson 2—from sample summaries to inference (12 points)

(a) The sample mean is

$$\bar{y} = \frac{5200}{100} = 52 \text{ points}.$$

The supplied raw-square identity gives

$$S_{YY} = 280300 - 100(52)^2 = 280300 - 270400 = 9900 \text{ points}^2.$$

Thus

$$s_Y^2 = \frac{9900}{100 - 1} = 100 \text{ points}^2, \qquad s_Y = 10 \text{ points}, \qquad SE(\bar{Y}) = \frac{10}{\sqrt{100}} = 1 \text{ point}.$$

The sample SD describes dispersion of individual scores; the SE describes sampling uncertainty in the estimated mean. Award 0.5 points for the mean, 0.75 for the centered sum of squares, 0.75 for the sample variance, 0.5 for the sample SD, and 0.5 for the SE, with appropriate units (3 points in total).

Lecture 2, PDF p. 8, "Sample Variance and Sample Standard Deviation":

"The sample variance, s_Y^2 is an estimator of the population variance σ_Y^2"

The page defines $s_Y^2 = (n-1)^{-1}\sum_i (Y_i - \overline{Y})^2$ and $s_Y = \sqrt{s_Y^2}$. Application: the raw-square identity in the question follows by expanding $(y_i - \bar{y})^2$ and using $\sum_i y_i = n\bar{y}$; it is an algebraic shortcut, not a sentence quoted from the lecture.

Lecture 2, PDF p. 10, "Step 2: Estimating the Standard Error":

"Replacing σ_Y by s_Y gives the estimated standard deviation of $\overline{Y}$:"

The displayed formula is $SE(\overline{Y}) = s_Y/\sqrt{n}$.

Lecture 2, PDF p. 10, "Step 2: Estimating the Standard Error":

"The standard error measures the sampling uncertainty of the estimator $\overline{Y}$."

(b) The hypotheses are $H_0 : \mu_Y = 50$ and $H_1 : \mu_Y \neq 50$. The observed statistic is

$$t_{\text{obs}} = \frac{52 - 50}{1} = 2.$$

At 5%, $|2| > 1.96$, so reject H_0. At 1%, $|2| < 2.576$, so do not reject H_0. This uses the stated large-sample normal approximation; no exact Student t reference distribution is being assumed. Award 0.5 points for each hypothesis, 1 point for the correctly centered and standardized statistic, and 0.5 for each correct critical-value comparison and decision (3 points in total).

Lecture 2, PDF p. 7, "P-Value for a One-Sample Mean Test":

"We test a population mean against a hypothesized value:"

The hypotheses on that page are $H_0 : \mu_Y = \mu_Y^0$ versus $H_1 : \mu_Y \neq \mu_Y^0$. Set $\mu_Y^0 = 50$ here. The statistic displayed on Lecture 2, PDF p. 13, "The t-statistic and its use for hypothesis testing", is $t = (\overline{y} - \mu_Y^0)/SE(\overline{Y})$.

Supplemental practice: Seminar 2, PDF p. 10, "Solution 2: Decisions and the meaning of .0164 (continued)":

"Reject H_0 when the p-value is at most α, or equivalently when $|t^{\text{obs}}|$ reaches the two-sided critical value."

The seminar table gives 1.960 at 5% and 2.576 at 1%. The table on Lecture 2, PDF p. 21 rounds the 1% value to 2.58; that rounding gives the same decision in this problem.

(c) The 95% normal approximate interval is

$$52 \pm 1.96(1) = [50.04, 53.96] \text{ points.}$$

It excludes 50, in agreement with rejection by the two-sided 5% test. Do not round the lower endpoint to 50 before checking inclusion. Award 0.75 points for the interval formula, 0.75 for each endpoint, and 0.75 for checking 50 and connecting the result to the test (3 points in total).

Lecture 2, PDF p. 21, "Computing CIs":

"Use $\overline{y} \pm c \times SE(\overline{Y})$, adding and subtracting c standard errors."

The same slide says: "Excluding 175 cm is equivalent to rejecting $H_0 : \mu_Y = 175$ cm at 5% (two-sided)." Application: replace the lecture's null value 175 by this problem's null value 50; it is excluded here.

(d) Using the planning SD, the planned half-width is

$$h(n) = 1.96\frac{10}{\sqrt{n}} = \frac{19.6}{\sqrt{n}}.$$

The required inequality is

$$\frac{19.6}{\sqrt{n}} \leq 0.98 \quad \Longleftrightarrow \quad \sqrt{n} \geq 20 \quad \Longleftrightarrow \quad n \geq 400.$$

The minimum integer is therefore 400 students. It is four times the original sample size. With the SD unchanged, quadrupling n halves the planned SE from 1 to 0.5 and the planned half-width from 1.96 to 0.98. The actual

interval width can differ if the new survey's estimated SD differs from the planning value. Award 1 point for the half-width inequality, 1 point for the minimum sample size, and 1 point for the sample-size/precision comparison (3 points in total).

Lecture 2, PDF p. 9, "Step 1: The Standard Deviation of the Sample Mean":

"This is smaller when n is larger: averaging more observations makes the sample mean more precise."

That page gives the theoretical SD $\sigma_Y/\sqrt{n}$. Combine it with the interval construction on Lecture 2, PDF p. 21 to form the half-width. This planning inequality is an application of those lecture formulas.

Supplemental practice: Seminar 2, PDF p. 20, "Solution 6: Sample size for a target half-width":

"For $m_n \leq m$, solve and round up:"

The seminar's formula is $n_{\min} = \lceil (1.96\sigma/m)^2 \rceil$, where m denotes target half-width. Insert $\sigma = 10$ and $m = 0.98$ to obtain exactly 400.

Question 14: Lesson 3—simple OLS from five observations (12 points)

(a) The means are $\bar{x} = 15/5 = 3$ and $\bar{y} = 35/5 = 7$. Set $d_{xi} = x_i - 3$ and $d_{yi} = y_i - 7$. The centered calculations are:

i	x_i	y_i	d_{xi}	d_{yi}	d_{xi}^2	$d_{xi}d_{yi}$
1	1	4	−2	−3	4	6
2	2	4	−1	−3	1	3
3	3	7	0	0	0	0
4	4	8	1	1	1	1
5	5	12	2	5	4	10
Sum	15	35	0	0	10	20

Therefore $S_{XX} = 10$, $S_{XY} = 20$, $\widehat{\beta}_1 = 20/10 = 2$ and $\widehat{\beta}_0 = 7 - 2(3) = 1$. The fitted line is $\widehat{y}_i = 1 + 2x_i$. One additional weekly study hour is associated with two additional predicted quiz points. This interpretation of the fitted association alone does not establish a causal effect. Award 0.5 points for each mean, 0.5 for each centered sum, 0.5 for each coefficient, 0.5 for the fitted line, and 0.5 for the slope interpretation.

Lecture 3, PDF p. 8, "The Ordinary Least Squares (OLS) Estimator":

"The intercept adjusts the average outcome for the fitted slope."

The same page gives

$$\hat{\beta}_1 = \frac{\sum_i (Y_i - \overline{Y})(X_i - \overline{X})}{\sum_i (X_i - \overline{X})^2}, \qquad \hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}.$$

Its other bullet says: "The slope compares the comovement of X and Y with the variation in X." Application: the centered table supplies the numerator and denominator directly; the slope is measured in quiz points per weekly study hour.

(b) Use $\widehat{y}_i = 1 + 2x_i$ and $\widehat{u}_i = y_i - \widehat{y}_i$:

i	x_i	y_i	$\widehat{y}_i$	$\widehat{u}_i$	$x_i\widehat{u}_i$
1	1	4	3	1	1
2	2	4	5	−1	−2
3	3	7	7	0	0
4	4	8	9	−1	−4
5	5	12	11	1	5
Sum	15	35	35	0	0

Thus $\sum_i \widehat{u}_i = 1-1+0-1+1 = 0$ and $\sum_i x_i\widehat{u}_i = 1-2+0-4+5 = 0$. These are the sample OLS fitting identities for an intercept and one regressor. Individual residuals can be positive, negative or zero even though both sums equal zero. Award 1 point for the five fitted scores, 1 point for the residuals, and 0.5 for each verified zero sum. The formulas on Lecture 3, PDF p. 9, "Back to the Linear Regression Model", are $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ and $\hat{U}_i = Y_i - \hat{Y}_i$.

Lecture 3, PDF p. 7, "The Ordinary Least Squares (OLS) Estimator":

"The first-order conditions are"

$$\sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \qquad \sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0.$$

Application: use the estimated coefficients in the first two formulas and add the residuals and the hour-weighted residuals to check the latter two.

Supplemental practice: Seminar 3, PDF p. 11, "Solution 3: Normal equations and squared errors (continued)":

"These sample fitting identities do not establish causality."

(c) The squared deviations are:

i	$(y_i - 7)^2$	$(\widehat{y}_i - 7)^2$	$\widehat{u}_i^2$
1	9	16	1
2	9	4	1
3	0	0	0
4	1	4	1
5	25	16	1
Sum	$TSS = 44$	$ESS = 40$	$SSR = 4$

The decomposition is $44 = 40 + 4$. Hence

$$R^2 = \frac{ESS}{TSS} = \frac{40}{44} = \frac{10}{11} = 1 - \frac{4}{44}, \qquad SER = \sqrt{\frac{SSR}{n-2}} = \sqrt{\frac{4}{5-2}} = \frac{2}{\sqrt{3}}.$$

The fitted line explains 10/11 of the sample variation in quiz scores; 1/11 remains in the residuals. SER is residual dispersion measured in quiz points, with two fitted coefficients accounted for by $n - 2$. It is not a standard error of a coefficient. Accept $\sqrt{4/3}$ or $2\sqrt{3}/3$ for the same exact SER. Award 0.5 points each for TSS, ESS and SSR, 0.75 for R^2 and 0.75 for SER. The displayed decomposition verifies the three sums; carry forward one consistent arithmetic error without penalizing it twice.

Lecture 3, PDF p. 13, "R^2":

"The R^2 is the ratio of the sample variance of $\hat{Y}$ to the sample variance of Y"

That page displays $R^2 = ESS/TSS = 1 - SSR/TSS$ with the corresponding squared sum definitions.

Lecture 3, PDF p. 16, "SER and RMSE: What Is the Difference?":

"The two degrees of freedom are used to estimate β_0 and β_1."

The SER formula on that page is $SER = \sqrt{SSR/(n-2)}$.

Supplemental practice: Seminar 3, PDF p. 19, "Solution 6: Decomposing the observed score variation":

"Thus $TSS = ESS + SSR$: $32 = 25.6 + 6.4$."

Application: the seminar demonstrates the same identity with different numbers. The exam calculations give $44 = 40+4$ and the exact residual spread $2/\sqrt{3}$; neither a decimal approximation nor a probability table is needed.

(d) Because $X_i = M_i/60$, substitute into the fitted line:

$$\widehat{y}_i = 1 + 2\left(\frac{M_i}{60}\right) = 1 + \frac{1}{30}M_i.$$

The intercept is still 1 quiz point. The slope is $1/30$ of a quiz point per additional study minute: 60 minutes correspond to the original two-point change per study hour. This is a change of regressor units, so every fitted score and residual stays the same. The outcome units, sample size and sums of squares also stay the same. Therefore $R^2 = 10/11$ and $SER = 2/\sqrt{3}$ quiz points remain unchanged. Award 0.5 points each for the intercept, minute slope, unchanged R^2 and unchanged SER, with the invariance explanation included in the last two items.

Supplemental practice: Seminar 3, PDF p. 13, "Solution 4: Shares, percentage points, and predicted scores":

"Changing the units changes the coefficient's numerical scale and preserves the predictions."

The seminar converts a share to percentage points. Application: here the same substitution method converts hours to minutes; $X_i = M_i/60$ is the reason the slope is divided by 60.

Lecture 3, PDF p. 16, "SER and RMSE: What Is the Difference?":

"Both are measured in the units of Y. They describe the typical size of the part of Y that the regression does not explain."

Application: the fitted scores, residuals and outcome units have not changed. Thus the sums in the R^2 definition on Lecture 3, PDF p. 13 and the SER definition on Lecture 3, PDF p. 16 remain the same. The numerical invariance after this specific hours-to-minutes conversion is a deduction, not a separate verbatim lecture example.